

DIFFERENTIAL GEOMETRY AND LIE GROUPS

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Fifth Semester

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Chapter 1

MANIFOLDS

July 20th.

The primary reference for this course will be Loring Tu's *An Introduction to Manifolds*. Prerequisites include Analysis of Several Variables, Linear Algebra, and Topology. The mid-term will account for 35% of the final grade, quizzes and student lectures will account for 15%, and the final exam will account for 50%.

Simply put, a manifold is any object that is locally Euclidean.

1.1 Calculus on $\mathbb{R}^n$

A point p or element in $\mathbb{R}^n$ is associated with an n -tuple of coordinates $x^1, x^2, \dots, x^n$. Let U be an open neighbourhood of p in $\mathbb{R}^n$. For a non-negative integer k , we say that a function $f : U \rightarrow \mathbb{R}$ is C^k -continuous at $p \in U$ if all partial derivatives of f of order $\leq k$ exist and are continuous at p . f is said to be C^∞ or *smooth* if f is C^k for all $k \geq 0$. f is C^k (or C^∞) on U if it is C^k (or C^∞) at every point in U .

If f happens to be a vector-valued function, i.e. $f : U \rightarrow \mathbb{R}^m$, we say $f = (f^1, f^2, \dots, f^m)$ is C^k (or C^∞) at p if each component function $f^i : U \rightarrow \mathbb{R}$ is C^k (or C^∞) at p . f is C^k (or C^∞) on U if it is C^k (or C^∞) at every point in U .

Definition 1.1. Let R be a commutative ring with unity. We term A to be an R -algebra if A is a commutative ring with unity and an R -module, and for $r \in R$ and $a, b \in A$, we have $r(ab) = (ra)b = a(rb)$.

In this course, we will be primarily be concerned with the case when R is a field, and A is an R -vector space.

For a subset $U \subseteq \mathbb{R}^n$, we denote $C^k(U)$ (or $C^\infty(U)$) as the set of all C^k (or C^∞) functions from U to $\mathbb{R}$. Note that this is a commutative ring with unity. It is also a vector space over $\mathbb{R}$. Together, $C^k(U)$ (or $C^\infty(U)$) is a (commutative) algebra over $\mathbb{R}$. Moreover, we have the descending chain of subalgebras

$$C^0(U) \supseteq C^1(U) \supseteq C^2(U) \supseteq \dots C^k(U) \supseteq C^{k+1}(U) \supseteq \dots \quad (1.1)$$

with

$$\bigcap_{k \geq 0} C^k(U) = C^\infty(U). \quad (1.2)$$

$f : U \rightarrow \mathbb{R}$ is termed *real analytic* at $p \in U$ if there exists a neighbourhood $V \subseteq U$ of p such that

$$f(x) = f(p) + \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p)(x^i - p^i) + \dots + \frac{1}{k!} \sum_{1 \leq i_1, \dots, i_k \leq n} \frac{\partial^k f}{\partial x^{i_1} \dots \partial x^{i_k}}(p)(x^{i_1} - p^{i_1}) \dots (x^{i_k} - p^{i_k}) + \dots \quad (1.3)$$

for all $x \in V$. f is said to be real analytic on U if it is real analytic at every point in U . We denote the set of all real analytic functions on U as $C^\omega(U)$. Note that $C^\omega(U) \subseteq C^\infty(U)$.

Theorem 1.2 (*Taylor's theorem*). Let $p \in U \subseteq \mathbb{R}^n$, and $f : U \rightarrow \mathbb{R}$ be a C^∞ function. Further assume that for all $x \in U$, the line segment from p to x is contained in U (U in this case is said to be star-shaped with respect to p). Then there exist $g_1, \dots, g_n \in C^\infty(U)$ such that, for all $x \in U$,

$$f(x) = f(p) + \sum_{i=1}^n g_i(x)(x^i - p^i), \quad (1.4)$$

with

$$g_i(p) = \frac{\partial f}{\partial x^i}(p), \quad 1 \leq i \leq n. \quad (1.5)$$

Proof. Let $x \in U$. Then the line segment $\{p + t(x - p) \mid t \in [0, 1]\}$ is contained in U . Composing f with the map $t \mapsto p + t(x - p)$, we obtain

$$\frac{df}{dt}(p + t(x - p)) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p + t(x - p))(x^i - p^i). \quad (1.6)$$

Integrating from 0 to 1, we get

$$f(x) - f(p) = \int_0^1 \frac{df}{dt}(p + t(x - p))dt = \sum_{i=1}^n (x^i - p^i) \int_0^1 \frac{\partial f}{\partial x^i}(p + t(x - p))dt = \sum_{i=1}^n (x^i - p^i)g_i(x). \quad (1.7)$$

Clearly, because f is C^∞ , each g_i is also C^∞ . Further, we have $g_i(p) = \int_0^1 \frac{\partial f}{\partial x^i}(p)dt = \frac{\partial f}{\partial x^i}(p)$. ■

Let $p = (p^1, \dots, p^n)$, $v \in \mathbb{R}^n$, and $f : U \rightarrow \mathbb{R}$ be C^∞ at p . The *directional derivative* of f at p in the direction of v is defined as

$$\lim_{t \rightarrow 0} \frac{f(p + tv) - f(p)}{t} = \left. \frac{d}{dt} f(p + tv) \right|_{t=0} = \sum_{i=1}^n \frac{df}{dx^i}(p)v^i. \quad (1.8)$$

From here, we obtain the operator

$$D_v = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i} \Big|_p \quad (1.9)$$

Now consider all pairs of the form (f, U) where $p \in U \subseteq \mathbb{R}^n$ is an open neighbourhood of p and $f : U \rightarrow \mathbb{R}$ is C^∞ . Define an equivalence relation $\sim$ on this set by declaring $(f, U) \sim (g, V)$ if and only if there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$. The equivalence class of (f, U) is denoted by $[(f, U)]$, and the set of all such equivalence classes is denoted by $C_p^\infty(\mathbb{R}^n)$. Such an equivalence class $[(f, U)]$ is called the *germ* of f at p . We can give $C_p^\infty(\mathbb{R}^n)$ the structure of a commutative ring with unity by defining

$$[(f, U)] + [(g, V)] = [(f + g, U \cap V)], \quad [(f, U)] \cdot [(g, V)] = [(fg, U \cap V)], \quad 1 = [(1, \mathbb{R}^n)]. \quad (1.10)$$

It can also be given the structure of a vector space over $\mathbb{R}$ by defining

$$\lambda[(f, U)] = [(\lambda f, U)], \quad \lambda \in \mathbb{R}. \quad (1.11)$$

Thus, $C_p^\infty(\mathbb{R}^n)$ is an $\mathbb{R}$ -algebra. Now, given a germ $\alpha = [(f, U)] \in C_p^\infty(\mathbb{R}^n)$ and a vector $v \in \mathbb{R}^n$, we define

$$D_v \alpha = \sum_{i=1}^n v^i \frac{\partial f}{\partial x^i} \Big|_p = v \cdot \nabla f(p). \quad (1.12)$$

Note that the choice of f is immaterial, since if $[(f, U)] = [(g, V)]$, then there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$, and hence $\nabla f(p) = \nabla g(p)$. One can infer that $D_{\lambda v}(f) = \lambda D_v(f) = D_v(\lambda f)$ and $D_v(f + g) = D_v(f) + D_v(g)$ for all $\lambda \in \mathbb{R}$ and $f, g \in C_p^\infty(\mathbb{R}^n)$. Thus, D_v is an $\mathbb{R}$ -linear map from $C_p^\infty(\mathbb{R}^n)$ to $\mathbb{R}$ (hereforth, we shall use f to also denote the germ $[(f, U)]$ of f at p). Also,

$$D_v(fg) = \sum_{i=1}^n v^i \frac{\partial(fg)}{\partial x^i}(p) = \sum_{i=1}^n v^i \left(f(p) \frac{\partial g}{\partial x^i}(p) + g(p) \frac{\partial f}{\partial x^i}(p) \right) = f(p)D_v(g) + g(p)D_v(f). \quad (1.13)$$

Such a property makes D_v what we call a derivation.

Definition 1.3. Any $\mathbb{R}$ -linear map $T : C_p^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}$ satisfying the Leibniz rule $T(fg) = f(p)T(g) + g(p)T(f)$ for all $f, g \in C_p^\infty(\mathbb{R}^n)$ is called a *derivation* at p .

One may show that if T is a derivation at p , then $T(1) = 0$.

The set of all derivations at p is denoted by $\mathfrak{D}_p(\mathbb{R}^n)$. It is easy to see that $\mathfrak{D}_p(\mathbb{R}^n)$ is a vector space over $\mathbb{R}$. Now consider the map

$$\theta : \mathbb{R}^n \rightarrow \mathfrak{D}_p(\mathbb{R}^n), \quad v \mapsto D_v. \quad (1.14)$$

This θ is an $\mathbb{R}$ -linear map (verify). We claim that θ is an isomorphism of real vector spaces. To this end, let $v \in \mathbb{R}^n$ such that $D_v = 0 \in \mathfrak{D}_p(\mathbb{R}^n)$. Then $D_v f = 0$ for all $f \in C_p^\infty(\mathbb{R}^n)$. In particular, for the coordinate functions x^i , we have $D_v x^i = v_i = 0$ for all $1 \leq i \leq n$. Thus, $v = 0$, and hence θ is injective. To show surjectivity, we appeal to Taylor's theorem; let $D \in \mathfrak{D}_p(\mathbb{R}^n)$. We claim that $D = D_v$ where $v = (D(x^1), \dots, D(x^n))$. Indeed, for any $f \in C_p^\infty(\mathbb{R}^n)$, we have

$$\begin{aligned} D(f) &= D(f(p)) + D\left(\sum_{i=1}^n (x^i - p^i)g_i(x)\right) = 0 + \sum_{i=1}^n ((x^i - p^i)(p)Dg_i(x) + D(x^i - p^i)g_i(p)) \\ &= \sum_{i=1}^n (Dx^i)g_i(p) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p)Dx^i = D_v(f). \end{aligned} \quad (1.15)$$

Hence, θ is surjective, and we have the desired isomorphism $\mathfrak{D}_p(\mathbb{R}^n) \cong \mathbb{R}^n$.

Note that a germ $f \in C_p^\infty(\mathbb{R}^n)$ has a valid value at p , but no other point. Hence, we have the evaluation map

$$\text{ev}_p : C_p^\infty(\mathbb{R}^n) \rightarrow \mathbb{R}, \quad [(f, U)] \mapsto f(p). \quad (1.16)$$

The kernel here is $\{[(f, U)] \in C_p^\infty(\mathbb{R}^n) \mid f(p) = 0\}$, which is an ideal in $C_p^\infty(\mathbb{R}^n)$. For any other element $[(f, U)] \in C_p^\infty(\mathbb{R}^n)$ with $f(p) \neq 0$, a valid inverse is given by $[(f^{-1}, U \setminus \{f = 0\})]$. This makes $C_p^\infty(\mathbb{R}^n)$ a local ring.

1.2 Multilinear Algebra

July 22nd.

Hereforth, V shall denote a finite-dimensional real vector space, and V^V shall denote its *dual space* $\text{Hom}_{\mathbb{R}}(V, \mathbb{R})$. If $e_1, \dots, e_n$ are a basis for V , then the *dual basis* of V^V is denoted by $\alpha^1, \dots, \alpha^n \in V^V$, and is defined by $\alpha^i(e_j) = \delta_{i,j}$ for all $1 \leq i, j \leq n$. One may verify that the dual basis $\{\alpha^1, \dots, \alpha^n\}$, induced by the basis $\{e_1, \dots, e_n\}$, is indeed a basis for V^V . S_k shall denote the permutation group on k symbols, satisfying $|S_k| = k!$. For any $\sigma \in S_k$, the sign of σ is denoted by $\text{sgn}(\sigma)$, and is defined as $+1$ if σ is an even permutation, and -1 if σ is an odd permutation.

With notation out of the way, let $k \geq 1$ be an integer. We shall consider the k -tuple $V^k = V \times \dots \times V$ (k times).

Definition 1.4. A mapping $f : V^k \rightarrow \mathbb{R}$ is said to be *k-linear* if f is linear in each of its k arguments.

A special case is when $k = 2$, in which case f is called a *bilinear form*. Such an $f : V \times V \rightarrow \mathbb{R}$ satisfies

$$f(\lambda v + \mu w, u) = \lambda f(v, u) + \mu f(w, u), \quad f(u, \lambda v + \mu w) = \lambda f(u, v) + \mu f(u, w). \quad (1.17)$$

A common example of a bilinear form is any valid inner product $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ on V . Another example is the determinant $\det : \mathbb{R}^n \rightarrow \mathbb{R}$, where the vectors are interpreted as the columns of a matrix. We can also write

$$L_k(V) = \{f : V^k \rightarrow \mathbb{R} \mid f \text{ is } k\text{-linear}\}. \quad (1.18)$$

It is easy to see that $L_k(V)$ is an $\mathbb{R}$ -vector space.

Definition 1.5. A k -linear map $f : V^k \rightarrow \mathbb{R}$ is called a *symmetric k -linear form* if for all $\sigma \in S_k$,

$$f(v_1, \dots, v_k) = f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \quad (1.19)$$

That is, f is invariant under any permutation of its arguments. Moreover, f is called an *alternating k -linear form* if for all $\sigma \in S_k$,

$$f(v_1, \dots, v_k) = \text{sgn}(\sigma) f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \quad (1.20)$$

One can see that the standard inner product on $\mathbb{R}^n$ is a symmetric bilinear form, and the determinant is an alternating n -linear form. Related to these definitions, we can immediately define the following subspaces of $L_k(V)$:

$$S_k(V) = \{f \in L_k(V) \mid f \text{ is symmetric}\} \subseteq L_k(V), \quad A_k(V) = \{f \in L_k(V) \mid f \text{ is alternating}\} \subseteq L_k(V). \quad (1.21)$$

Currently, our main interest of study is $A_k(V)$, the space of alternating k -linear forms on V . Elements of $A_k(V)$ are termed *alternating k -tensors*. Similarly, those of $S_k(V)$ and $L_k(V)$ are called *symmetric k -tensors* and *k -tensors*, respectively. For $k = 0$, we define $L_0(V) = S_0(V) = A_0(V) = \mathbb{R}$. For $k = 1$, we have $L_1(V) = S_1(V) = A_1(V) = V^V$. We can define the (left) action of S_k on $L_k(V)$ by

$$(\sigma \cdot f)(v_1, \dots, v_k) = f(v_{\sigma(1)}, \dots, v_{\sigma(k)}). \quad (1.22)$$

Thus, f is symmetric if $\sigma \cdot f = f$ for all $\sigma \in S_k$, and f is alternating if $\sigma \cdot f = \text{sgn}(\sigma)f$ for all $\sigma \in S_k$.

Given an $f \in L_k(V)$, we can define the *symmetrization* $Sf \in S_k(V)$, of f , as

$$Sf = \sum_{\sigma \in S_k} \sigma \cdot f. \quad (1.23)$$

One may, indeed, check that Sf is symmetric. Similarly, we can define the *alternation* $Af \in A_k(V)$, of f , as

$$Af = \sum_{\sigma \in S_k} \text{sgn}(\sigma) \sigma \cdot f. \quad (1.24)$$

1.2.1 The Tensor and Wedge Products

July 24th.

For any $f \in L_k(V)$ and $g \in L_l(V)$, we can define the *tensor product* $f \otimes g \in L_{k+l}(V)$ as

$$(f \otimes g)(v_1, \dots, v_{k+l}) = f(v_1, \dots, v_k)g(v_{k+1}, \dots, v_{k+l}). \quad (1.25)$$

Verify that $f \otimes g$ is indeed $(k+l)$ -linear. We are interested in the tensor product of alternating tensors, so let $f \in A_k(V)$ and $g \in A_l(V)$. Then, we can define the *wedge product* $f \wedge g \in A_{k+l}(V)$ as

$$f \wedge g = \frac{1}{k!l!} A(f \otimes g) = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \cdot (f \otimes g). \quad (1.26)$$

The associativity of the tensor product holds easily: for $f \in L_k(V)$, $g \in L_l(V)$, and $h \in L_m(V)$,

$$(f \otimes g) \otimes h = f \otimes (g \otimes h). \quad (1.27)$$

Proposition 1.6. For $f \in A_k(V)$ and $g \in A_l(V)$, we have

$$f \wedge g = (-1)^{kl} g \wedge f. \quad (1.28)$$

Proof. We shall show that $A(f \otimes g) = (-1)^{kl} A(g \otimes f)$. To this end, fix the permutation $\tau \in S_{k+l}$ defined by

$$\tau = \begin{pmatrix} 1 & 2 & \cdots & l & l+1 & l+2 & \cdots & l+k \\ k+1 & k+2 & \cdots & k+l & 1 & 2 & \cdots & k \end{pmatrix}. \quad (1.29)$$

We have

$$\begin{aligned}
A(f \otimes g)(v_1, \dots, v_{k+l}) &= \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) f(v_{\sigma(1)}, \dots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)}) \\
&= \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) f(v_{\sigma\tau(l+1)}, \dots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \dots, v_{\sigma\tau(l)}) \\
&= \text{sgn}(\tau) \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma\tau) f(v_{\sigma\tau(l+1)}, \dots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \dots, v_{\sigma\tau(l)}) \\
&= \text{sgn}(\tau) A(g \otimes f)(v_1, \dots, v_{k+l}).
\end{aligned} \tag{1.30}$$

With $\text{sgn}(\tau) = (-1)^{kl}$, the proof is complete. $\blacksquare$

Corollary 1.7. For $f \in A_k(V)$, we have $f \wedge f = 0$ if k is odd.

Proof. Easy since $f \wedge f = (-1)^{k^2} f \wedge f = -f \wedge f$ if k is odd, so $f \wedge f = 0$. $\blacksquare$

Lemma 1.8. For $f \in L_k(V)$ and $g \in L_l(V)$, we have

$$A(Af \otimes g) = k!A(f \otimes g), \quad A(f \otimes Ag) = l!A(f \otimes g). \tag{1.31}$$

Proof. We shall prove the first equality; the second follows similarly. We have

$$\begin{aligned}
A(Af \otimes g) &= \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \cdot \left(\sum_{\tau \in S_k} \text{sgn}(\tau) (\tau \cdot f) \otimes g \right) = \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \cdot \left(\sum_{\tau \in S_k} \text{sgn}(\tau) \tau \cdot (f \otimes g) \right) \\
&= \sum_{\tau \in S_k} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma\tau) \sigma\tau \cdot (f \otimes g) = k! \sum_{\alpha \in S_{k+l}} \text{sgn}(\alpha) \alpha \cdot (f \otimes g) = k!A(f \otimes g).
\end{aligned} \tag{1.32}$$

Proposition 1.9. The wedge product is associative, i.e. for $f \in A_k(V)$, $g \in A_l(V)$, and $h \in A_m(V)$, we have

$$(f \wedge g) \wedge h = f \wedge (g \wedge h). \tag{1.33}$$

Proof. We have

$$\begin{aligned}
(f \wedge g) \wedge h &= \frac{1}{(k+l)!m!} A((f \wedge g) \otimes h) = \frac{1}{(k+l)!m!k!l!} A(A(f \otimes g) \otimes h) = \frac{(k+l)!}{(k+l)!l!k!m!} A((f \otimes g) \otimes h) \\
&= \frac{1}{k!l!m!} A(f \otimes g \otimes h) = f \wedge (g \wedge h).
\end{aligned} \tag{1.34}$$

Corollary 1.10. $f \wedge g \wedge h = \frac{1}{k!l!m!} A(f \otimes g \otimes h)$ for $f \in A_k(V)$, $g \in A_l(V)$, and $h \in A_m(V)$. Inductively,

$$f_1 \wedge \dots \wedge f_k = \frac{1}{k_1! \dots k_k!} A(f_1 \otimes \dots \otimes f_k), \tag{1.35}$$

where $f_i \in A_{k_i}(V)$ for $1 \leq i \leq k$.

Proposition 1.11. Let $\alpha^1, \dots, \alpha^n \in V^V = A_1(V) = L_1(V)$, and $v_1, \dots, v_k \in V$. Then

$$(\alpha^1 \wedge \dots \wedge \alpha^k)(v_1, \dots, v_k) = \det(\alpha^i(v_j))_{1 \leq i, j \leq k}. \tag{1.36}$$

Proof. We have

$$\begin{aligned}
 (\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \dots, v_k) &= \sum_{\sigma \in S_k} \text{sgn}(\sigma) (\alpha^1 \otimes \cdots \otimes \alpha^k)(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \\
 &= \sum_{\sigma \in S_k} \text{sgn}(\sigma) \alpha^1(v_{\sigma(1)}) \alpha^2(v_{\sigma(2)}) \cdots \alpha^k(v_{\sigma(k)}) = \det(\alpha^i(v_j))_{1 \leq i, j \leq k}. \quad (1.37)
 \end{aligned}$$

■

Let V be an $\mathbb{R}$ -vector space with basis $e_1, \dots, e_n$, and V^V be its dual space with dual basis $\alpha^1, \dots, \alpha^n$. Moreover, let I be a multi-index of length k , i.e. $I = (i_1, \dots, i_k)$ with $1 \leq i_1, \dots, i_k \leq n$. We define $e_I := (e_{i_1}, \dots, e_{i_k}) \in V^k$ and $\alpha^I := \alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k} \in A_k(V)$. Then, a k -linear form $f \in L_k(V)$ is completely determined by its values on the e_I , and an alternating k -linear form $f \in A_k(V)$ is completely determined by its values on the e_I with $i_1 < i_2 < \cdots < i_k$.

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